Closed-Loop PID Control of Light Intensity Using an LDR and WS2812 LED

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***Abstract*—This paper presents the design, implementation, and evaluation of a closed-loop PID control system for regulating light intensity using an LDR sensor and a WS2812 LED actuator. The system is implemented on an Arduino MKR1000 and incorporates advanced features including signal filtering, anti-windup mechanisms, rate limiting, and automatic calibration. The behaviour of both open-loop and closed-loop control systems is analysed and compared. Experimental results demonstrate that the closed-loop system provides significantly improved stability, accuracy, and disturbance rejection.**

I. Introduction

Control systems are fundamental in engineering applications where maintaining a desired output in the presence of disturbances is required. Open-loop systems, while simple, lack the ability to adapt to environmental changes. In contrast, closed-loop systems utilise feedback to continuously correct their output, resulting in improved performance.

This project implements a closed-loop light control system using an LDR sensor to measure ambient light and a WS2812 LED to adjust illumination. A PID controller is used to regulate the system, and its performance is analysed in comparison to open-loop control.

II. System Overview

As shown in Fig. 1, the implemented system forms a standard feedback loop in which the setpoint is compared with the measured light contribution. The PID controller converts the resulting error into a brightness command for the WS2812 LED. The emitted light is then sensed by the LDR, producing the feedback signal that closes the control loop.

Setpoint → PID Controller → LED → Light Output → LDR Sensor → Feedback

Fig. 1. Closed-loop control system block diagram.

III. Applications and System Context

The implemented system has direct relevance to a range of real-world applications. In smart lighting systems, feedback-based control is used to maintain consistent illumination levels while minimising energy consumption. Similarly, mobile devices such as smartphones and laptops employ ambient light sensors to dynamically adjust screen brightness, improving both usability and battery efficiency.

From an engineering perspective, the system demonstrates how feedback control enables adaptability in dynamic environments. However, this adaptability comes at the cost of increased system complexity and the requirement for careful tuning and signal conditioning. The selection of an LDR and WS2812 LED provides a simple yet effective platform for demonstrating these principles, although the nonlinearity of the sensor and noise sensitivity introduce additional challenges that must be addressed.

IV. Sensor Data Analysis

The LDR operates as a resistive sensor whose resistance decreases with increasing light intensity. When configured in a voltage divider, this produces an analogue voltage which is sampled by the Arduino's 10-bit ADC over the range 0 to 1023. This finite resolution introduces quantisation noise, which is small but still relevant when the controller is operating close to the setpoint.

In earlier implementations, a dual-sampling approach was used at every control step, where the LED was briefly turned off to measure ambient light, followed by a second measurement with the LED enabled. While accurate, this caused visible flicker due to the LED being repeatedly switched off.

To address this, the final implementation introduces an ambient light estimation strategy. Instead of measuring ambient light every cycle, the system maintains an estimate of ambient light:

$$PV = \text{Total Light} - \text{Ambient Estimate} \tag{1}$$

The ambient estimate is updated periodically using a slower refresh process:

$$A_{est} = A_{est} + \beta(A_{meas} - A_{est}) \tag{2}$$

where β is the ambient estimation gain.

This significantly reduces flicker because the LED remains continuously on during normal operation. However, a small disturbance is still introduced during periodic ambient refresh, as the LED must briefly be turned off to obtain a new ambient measurement.

To further improve signal quality, exponential filtering is applied:

$$PV_f = PV_f + \alpha(PV - PV_f) \tag{3}$$

This reduces measurement noise and stabilises the derivative term of the controller, at the cost of introducing a small delay. As shown in Fig. 2, the filtered signal is substantially smoother than the raw sensor signal during both transient and steady-state operation.

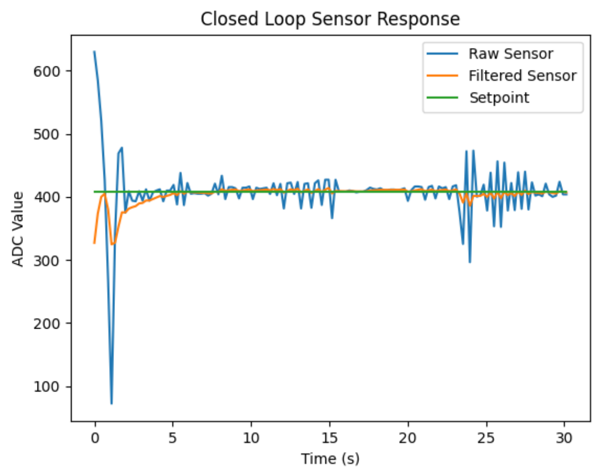

Fig. 2. Closed-loop sensor response showing raw sensor readings, filtered signal, and setpoint.

The raw sensor signal exhibits noticeable noise and occasional spikes, particularly during transient conditions and disturbances.

The filtered signal significantly reduces this noise through exponential smoothing, providing a stable signal for feedback control. The filtered output closely tracks the setpoint, demonstrating that the control system is effectively regulating the measured light intensity.

This figure highlights the importance of signal conditioning, as unfiltered sensor data would introduce instability, particularly in the derivative component of the controller.

V. Actuator Behaviour

The WS2812 LED is a digitally controlled actuator with 8-bit resolution, allowing brightness values between 0 and 255. Its response is effectively instantaneous compared to the system sampling rate, making it suitable for control applications.

However, the LED serves a dual role in this system: it is both the actuator and a contributor to the measured signal. This creates a coupling between actuation and sensing, which must be carefully managed.

The actuator also introduces a nonlinear input-output relationship. Although the controller commands an 8-bit brightness value, the perceived brightness seen by a human observer does not vary linearly because human vision has an approximately logarithmic response to light intensity. In addition, the PWM-based LED brightness command does not translate directly into a linear ADC response at the LDR, since the optical path, LED emission characteristics, and sensor response all introduce nonlinearity. This matters for control stability because a fixed increase in actuator command can produce different changes in measured output at different operating points. As a result, gains that appear acceptable in one region may become too aggressive or too weak elsewhere, affecting overshoot, settling behaviour, and visible flicker.

To prevent instability and visible flicker, a rate limiter is applied to the output:

$$u_k = \text{limit}(u_{k-1} \pm \Delta u) \tag{4}$$

where Δu is the maximum allowable change per control step.

Additionally, output saturation limits are enforced to ensure the actuator remains within valid bounds:

$$0 \leq u \leq 255 \tag{5}$$

A key limitation of this design is that ambient light must occasionally be measured by temporarily turning off the LED. Although this occurs infrequently, it introduces a small periodic disturbance, which is perceived as slight flicker. This represents a fundamental limitation of using a single sensor for both ambient and controlled light measurement.

VI. Open-Loop Control Behaviour

In the open-loop configuration, the LED brightness is fixed and does not respond to sensor input. As a result, the system cannot compensate for changes in ambient light.

Experimental observations show that while the sensor value varies significantly under changing conditions, the actuator output remains constant. This leads to poor performance in dynamic environments, as the system cannot maintain a consistent light level. As shown in Fig. 3, the sensed output changes while the commanded LED level remains fixed.

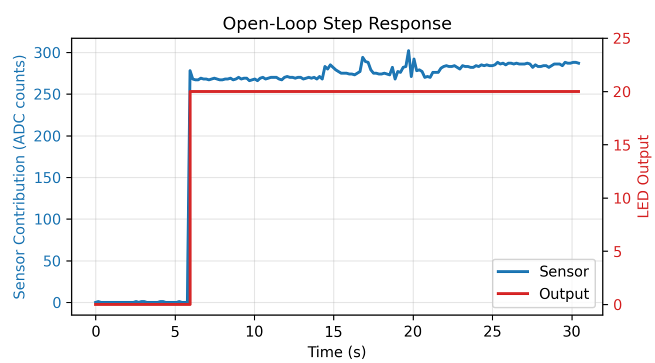

Fig. 3. Open-loop response showing constant output despite changing sensor input.

This demonstrates that open-loop control is insufficient for applications where environmental conditions are unpredictable.

VII. Closed-Loop PID Control

The system implements a discrete-time PID controller to regulate light intensity:

$$u(t) = K_p e(t) + K_i \int e(t)\,dt + K_d \frac{de(t)}{dt} \quad (6)$$

where the error is defined as:

$$e(t) = \text{Setpoint} - PV_f \quad (7)$$

The proportional term provides immediate correction, the integral term removes steady-state error, and the derivative term anticipates future changes by observing the rate of change of the measured signal.

To reduce noise sensitivity, the derivative is calculated on the filtered measurement rather than the raw error:

$$d = -K_d \frac{d(PV_f)}{dt} \quad (8)$$

This improves stability in the presence of sensor noise.

A. Advanced Control Enhancements

Several practical enhancements have been implemented to improve controller performance and robustness.

An integral active band is used to limit when the integral term is allowed to accumulate. This prevents excessive integral action when the system is far from the setpoint:

$$|e(t)| < E_{band} \quad (9)$$

An anti-windup mechanism prevents the integral term from growing when the actuator is saturated. This ensures faster recovery and avoids large overshoots.

Additionally, integral bleed is applied when the system crosses the setpoint:

$$I \leftarrow \gamma I \quad (10)$$

This reduces oscillations and improves settling behaviour.

A deadband is introduced around the setpoint:

$$|e(t)| < \epsilon \Rightarrow e(t) = 0 \quad (11)$$

This eliminates unnecessary control activity and reduces output flicker. As shown in Fig. 4, the control error decreases rapidly during the transient response while the actuator output settles to a bounded operating level.

At the beginning of the response, the error is large due to the difference between the measured value and the setpoint.

The controller responds by increasing the actuator output, which causes the error to decrease rapidly. As the system approaches the setpoint, the error converges towards zero, indicating successful regulation.

The controller output stabilises at a steady value, with small fluctuations corresponding to minor disturbances and sensor noise. Occasional spikes in the error signal are caused by external disturbances and ambient light estimation updates, which are quickly corrected by the controller.

Finally, the ambient estimation strategy allows the controller to operate continuously without frequent LED interruptions, significantly improving perceived stability while still maintaining accurate control.

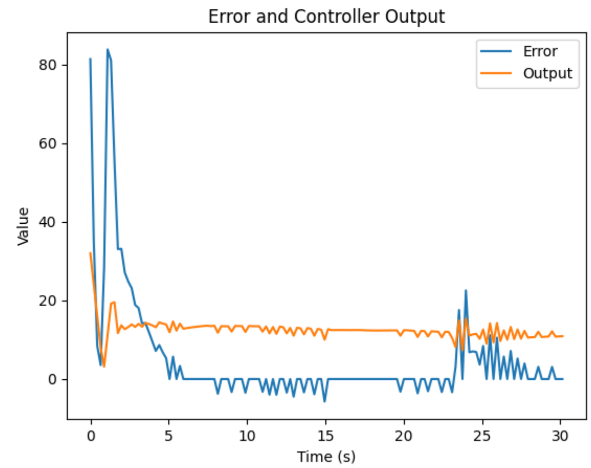

Fig. 4. Error signal and controller output over time during closed-loop operation.

B. System Behaviour

The closed-loop system demonstrates stable convergence to the desired setpoint. During startup, the controller rapidly increases the LED brightness until the measured contribution approaches the setpoint.

A small overshoot is observed due to the combined effect of proportional and integral action, after which the system settles to a steady operating point.

As shown in Fig. 5, the response exhibits a short rise time as the measured output approaches the setpoint, followed by a modest settling time before the output stabilises. The steady-state error remains close to zero because the integral term removes residual offset, while the final overshoot remains small due to the tuned proportional, integral, and derivative actions together with output limiting.

When external disturbances are introduced, such as additional light shining on the LDR, the controller compensates by reducing LED brightness. Conversely, when ambient light decreases, the controller increases output to maintain the desired level.

A minor periodic fluctuation can be observed in the output signal. This is caused by the ambient light refresh process, during which the LED is briefly turned off to update the ambient estimate. This introduces a small disturbance into the system, which is quickly corrected by the controller.

This behaviour highlights a practical limitation of the system: using a single sensor for both ambient and controlled light measurement inherently introduces a trade-off between measurement accuracy and actuator stability.

Initially, the measured value deviates significantly from the setpoint, resulting in a large control error. The controller rapidly adjusts the actuator output, driving the system towards the desired operating point.

A transient response is observed, including a brief overshoot and rapid correction. Following this, the system settles and

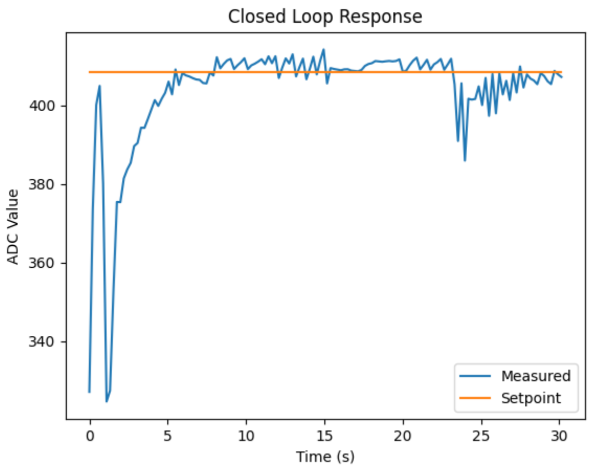

Fig. 5. Closed-loop response showing convergence of the measured output to the setpoint.

maintains operation close to the setpoint with minimal steady-state error.

Later fluctuations in the signal correspond to disturbances, such as external light interference and ambient recalibration. The controller successfully compensates for these disturbances, demonstrating effective disturbance rejection and stable closed-loop performance.

VIII. Design Methodology and Evaluation

The system was developed iteratively, starting with a simple open-loop implementation and progressing towards a fully featured closed-loop controller.

Initial implementations used direct dual-sampling at every control step, which provided accurate measurements but resulted in unacceptable flicker. This led to the development of the ambient estimation approach, which significantly improved visual stability.

PID gains were tuned experimentally to balance responsiveness and stability. Additional enhancements, including filtering, anti-windup, integral limiting, and rate limiting, were introduced to address specific observed issues such as overshoot, noise sensitivity, and actuator saturation.

As a result, filtering, anti-windup mechanisms, and rate limiting were incorporated into the design. Automatic calibration was also introduced to improve adaptability. Each of these enhancements was validated experimentally, leading to a progressively more robust system.

This iterative development process highlights the importance of testing and refinement in control system design. Rather than relying solely on theoretical models, practical implementation revealed additional challenges that required engineering solutions.

IX. Use of Scientific Principles

The project applies fundamental principles of control theory, including feedback systems, PID control, and signal processing. These principles were used to solve a practical engineering problem involving real-world constraints such as noise and actuator limitations.

X. Information Sources

The development of this system was informed by a combination of academic literature on control systems, technical documentation for hardware components, and experimental data obtained during testing. This combination of theoretical and practical sources enabled an effective and well-informed design process.

XI. Source Code and Demonstration

The full implementation is available at:

GitHub Repository:

https://github.com/Thomriiii/Sensors-Actuators-Controllers/tree/main/Coursework1

A demonstration of the system operation can be viewed at:

YouTube Video:

https://youtu.be/sudIHZlDrpw

XII. Conclusion

A closed-loop PID control system for regulating light intensity has been successfully implemented and evaluated. The system demonstrates clear improvements over open-loop control, including accurate setpoint tracking and effective disturbance rejection.

Advanced features such as filtering, anti-windup, and ambient estimation significantly enhance system performance and usability. In particular, the ambient estimation approach reduces visible flicker while maintaining acceptable measurement accuracy.

The main limitation of the final design is that a single LDR is used for both ambient estimation and controlled-light measurement. This creates measurement-actuation coupling, because the actuator state must briefly influence the sensing process whenever the ambient estimate is refreshed. Future work could reduce this disturbance by introducing a separate ambient sensor or by improving optical isolation between the LED and the LDR. Overall, the project demonstrates the practical application of control theory in a real-world system, including the challenges of noise, nonlinearity, and hardware constraints.